Practical No. 1

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Batch: B2 **Div: B**

Class: BE COMPUTER

Title:

Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyse their time and space complexity.

Program:

```
#include <bits/stdc++.h>
using namespace std;

int fibonacci(int n) {
    if (n <= 1) {
        return n;
    }
    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
    int i, n, t1 = 0, t2 = 1, nT, p;
    cout << "Fibonacci Series in C++ Without Using Recursion";
    cout << "\nEnter any number:";
    cin >> n;

    for (i = 1; i <= n; ++i) {
        cout << t1 << " ";
        nT = t1 + t2;
        t1 = t2;
        t2 = nT;
    }

    cout << "\n\nFibonacci Series in C++ Using Recursion";
    cout << "\nEnter any number:";
    cin >> p;
    cout << fibonacci(p);
}
```

Output:

```
(base) sspm@sspm:~$ cd Desktop
(base) sspm@sspm:~/Desktop$ g++ daa1.cpp
(base) sspm@sspm:~/Desktop$ ./a.out

Fibonacci Series in C++ Without Using Recursion
Enter any number:10
0 1 1 2 3 5 8 13 21 34

Fibonacci Series in C++ Using Recursion
Enter any number:10
55
```